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Reply-to: Matthias Kretz <m.kretz@gsi.de>
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Target: C++26

STD::SIMD IS A RANGE

ABSTRACT

P1928 “std::simd — merge data-parallel types from the Parallelism TS 2” promised a paper on making simd a range. This paper explores the addition of iterators to basic_simd and basic_simd_mask.

CONTENTS

1	CHANGELOG	1
1.1	CHANGES FROM REVISION 0	1
1.2	CHANGES FROM REVISION 1	1
2	STRAW POLLS	1
2.1	SG9 AT WROCŁAW 2024	1
3	INTRODUCTION, OR WHY SIMD WASN’T A RANGE IN THE TS	2
4	MOTIVATION	2
5	INTEGRATION WITH THE STANDARD LIBRARY	2
5.1	READ-ONLY SUBSCRIPT SHOULD IMPLY READ-ONLY ITERATION	3
5.2	PRESENT A RANGE OF SIMD AS A RANGE OF SIMD’S VALUE-TYPE	3
6	DOWNSIDES OF MAKING SIMD A RANGE	3
7	DESIGN CHOICE: SENTINEL	4
8	WORDING	4
A	BIBLIOGRAPHY	7

1

CHANGELOG

1.1

CHANGES FROM REVISION 0

Previous revision: P3480R0

- Simplify to a single iterator class template.
- Remove incorrect `operator-` overload.
- Discuss design choice of using a sentinel type for `end()`.

1.2

CHANGES FROM REVISION 1

Previous revision: P3480R1

- Add SG9 poll results.
- Use `default_sentinel_t` instead of a new sentinel type.
- Use an almost-mutable iterator type as directed by SG9 for non-const `begin()`.
- Fix `for_each` example to use `ranges` version.

2

STRAW POLLS

2.1

SG9 AT WROCŁAW 2024

Poll: We want `std::basic_simd` to be a range.

SF	F	N	A	SA
6	2	0	0	0

Poll: We want `std::basic_simd` to be a common range.

SF	F	N	A	SA
0	0	3	4	1

Poll: We want `std::basic_simd::operator[]` and `std::basic_simd::begin/end` in C++26 without mutation support, knowing that we might not be able to do it later due to ABI issues (e.g. `decltype(auto) f(std::simd<float> x) { return x[0]; }` could change return type).

SF	F	N	A	SA
6	2	0	0	0

Poll: We want `std::basic_simd::iterator` and `std::basic_simd::const_iterator` to be different types to make the transition to mutable iteration easier. This also means adding a non-const `begin()`

overload that returns a different type than the `const begin()` overload but currently has the same semantics.

SF	F	N	A	SA
2	3	3	0	0

Poll: Use `std::default_sentinel_t` instead of *simd-iterator-sentinel*.

→ unanimous consent

Poll: Forward P3480R1 with the changes above to LEWG for inclusion in C++26.

SF	F	N	A	SA
7	1	0	0	0

3

INTRODUCTION, OR WHY SIMD WASN'T A RANGE IN THE TS

The Parallelism TS 2 was based on C++17. Ranges were added in C++20. Before ranges, an iterator category was tied to whether `operator*` of iterators returned an lvalue reference. Since `basic_simd` and `basic_simd_mask` objects are not composed of sub-objects (in other words, a `simd<int>` contains no *int objects*), `operator[]` returns prvalues (or a proxy reference in the TS for the non-const case). An iterator needs to do the same and thus never could be in any other iterator category than *Cpp17InputIterator*. In reality, the iterator category always was “random access” (never contiguous; because while `basic_simd` is a contiguous range in memory it isn't one in the object model of C++). In order to not cement that mismatch, it was never proposed to make `basic_simd/basic_simd_mask` a range for the TS.

Now that the iterator concepts don't require an lvalue reference anymore we can easily make `basic_simd/basic_simd_mask` a read-only range. Iterator dereference would return a prvalue (a copy of the value stored in the `basic_simd/basic_simd_mask` object). In addition, the abstraction of a sentinel instead of an iterator pointing beyond the last value of the `basic_simd` seems like a useful tool for `basic_simd`.

4

MOTIVATION

After the technical reasons for *not* adding iterators to `basic_simd/basic_simd_mask` are resolved, we still need to consider why `basic_simd` should be a range in the first place.

5

INTEGRATION WITH THE STANDARD LIBRARY

We can improve integration of `basic_simd/basic_simd_mask` with the rest of the standard library. By making `basic_simd/basic_simd_mask` a range many of the existing facilities in the standard li-

brary become easily accessible. All of these facilities do work as intended — in other words: presenting `basic_simd/basic_simd_mask` as a range matches on the semantic level, not only syntactically.

5.1

READ-ONLY SUBSCRIPT SHOULD IMPLY READ-ONLY ITERATION

With P1928R12 we can write

```
std::simd<int> v = ...;
for (int i = 0; i < v.size(); ++i) {
    do_something(v[i]);
}
```

Why then, can we not also write

```
for (auto x : v) {
    do_something(x);
}
```

and

```
std::ranges::for_each(v.begin(), v.end(), [](auto x) {
    do_something(x);
});
```

and

```
v | std::views::filter([](auto x) { return x > 0; }) | std::ranges::to<std::vector>();
```

C++ users have learned that whenever a for loop with subscript does what they need to do, then a ranged for loop, standard algorithm, or range adaptor are valid alternatives. This expectation should not get an exception with `basic_simd` and `basic_simd_mask`.

5.2

PRESENT A RANGE OF SIMD AS A RANGE OF SIMD'S VALUE-TYPE

In some applications it is more efficient (and simpler) to work with `basic_simd` objects internally, instead of constantly doing loads and stores. Thus a fairly simple container that comes up in applications could be `std::vector<std::simd<float>>`. On I/O such an application typically cannot communicate in `basic_simd` objects anymore. Instead it needs to present a range of floats. Read-only iterators on `basic_simd` do not help with the input side. But for output we can easily turn the `vector<simd<float>>` into a range of float:

```
std::vector<std::simd<float>> data;
auto range_of_float = data | std::views::join;
```

6

DOWNSIDES OF MAKING SIMD A RANGE

Really, I can't think of any downsides of making `basic_simd/basic_simd_mask` a range. In principle one could argue that `basic_simd/basic_simd_mask` is not a container [P0851R0]. Consequently, it

shouldn't have a container interface and thus no iterators. But then we should probably remove the subscript operator as well.

7

DESIGN CHOICE: SENTINEL

The `basic_simd` iterator type must have a reference/pointer to the `basic_simd` object it is iterating together with an offset, where into the `basic_simd` it is pointing. Because of these two members (and their type), the iterator already knows the complete bounds of the range it is pointing into. Consequently, a single `basic_simd` iterator can always determine whether it points at the beginning or end of the range, it doesn't need to compare against another offset. A sentinel type allows asking that question via `operator==`. Thus, instead of comparing two runtime offset members on `operator==`, a compare against a sentinel is implemented as a compare against a compile-time constant. This makes it easier for the compiler to optimize and reduces the size of the `end()` sentinel to a single byte (empty type).

8

WORDING

This is just a sketch derived from my implementation of `basic_simd` and `basic_simd_mask` iterators. Add the following:

```
namespace std
{
    template <typename V>
        class simd-iterator           // exposition only
        {
            V* data_ = nullptr;        // exposition only
            int offset_ = 0;           // exposition only

        public:
            using value_type = typename V::value_type;
            using iterator_category = std::random_access_iterator_tag;
            using difference_type = int;

            constexpr simd-iterator() = default;

            constexpr
            simd-iterator(V& d, int x)
            : data_(&d), offset_(x)
            {}

            constexpr
            simd-iterator(const simd-iterator &) = default;
```

[simd.iterators]

```

constexpr simd-iterator&
operator=(const simd-iterator &) = default;

constexpr value_type
operator*() const
{ return (*data_)[offset_]; }

constexpr simd-iterator&
operator++()
{
    ++offset_;
    return *this;
}

constexpr simd-iterator
operator++(int)
{
    simd-iterator r = *this;
    ++offset_;
    return r;
}

constexpr simd-iterator&
operator--()
{
    --offset_;
    return *this;
}

constexpr simd-iterator
operator--(int)
{
    simd-iterator r = *this;
    --offset_;
    return r;
}

constexpr difference_type
operator-(simd-iterator rhs) const
{ return offset_ - rhs.offset_; }

constexpr friend difference_type
operator-(simd-iterator it, default_sentinel_t)
{ return it.offset_ - difference_type(V::size.value); }

```

```

constexpr friend difference_type
operator-(default_sentinel_t, simd_iterator it)
{ return difference_type(V::size.value) - it.offset_; }

constexpr friend simd_iterator
operator+(difference_type x, const simd_iterator& it)
{ return simd_iterator(*it.data_, it.offset_ + x); }

constexpr friend simd_iterator
operator+(const simd_iterator& it, difference_type x)
{ return simd_iterator(*it.data_, it.offset_ + x); }

constexpr friend simd_iterator
operator-(const simd_iterator& it, difference_type x)
{ return simd_iterator(*it.data_, it.offset_ - x); }

constexpr simd_iterator&
operator+=(difference_type x)
{
    offset_ += x;
    return *this;
}

constexpr simd_iterator&
operator-=(difference_type x)
{
    offset_ -= x;
    return *this;
}

constexpr value_type
operator[](difference_type i) const
{ return (*data_)[offset_ + i]; }

constexpr friend auto operator<=>(simd_iterator a, simd_iterator b)
{ return a.offset_ <=> b.offset_; }

constexpr friend bool operator==(simd_iterator a, simd_iterator b) = default;

constexpr friend bool operator==(simd_iterator a, default_sentinel_t)
{ return a.offset_ == difference_type(V::size.value); }
};
}

```

[simd.overview]

```
template<class T, class Abi> class basic_simd {
public:
    using value_type = T;
    using mask_type = basic_simd_mask<sizeof(T), Abi>;
    using abi_type = Abi;
    using iterator = simd-iterator<basic_simd>;
    using const_iterator = simd-iterator<const basic_simd>;

    constexpr iterator begin();
    constexpr const_iterator begin() const;
    constexpr const_iterator cbegin() const;
    constexpr default_sentinel_t end() const;
    constexpr default_sentinel_t cend() const;
```

[simd.mask.overview]

```
template<size_t Bytes, class Abi> class basic_simd_mask {
public:
    using value_type = bool;
    using abi_type = Abi;
    using iterator = simd-iterator<basic_simd_mask>;
    using const_iterator = simd-iterator<const basic_simd_mask>;

    constexpr iterator begin();
    constexpr const_iterator begin() const;
    constexpr const_iterator cbegin() const;
    constexpr default_sentinel_t end() const;
    constexpr default_sentinel_t cend() const;
```

A

BIBLIOGRAPHY

- [P0851R0] Matthias Kretz. *P0851R0: simd<T> is neither a product type nor a container type*. ISO/IEC C++ Standards Committee Paper. 2017. URL: <https://wg21.link/p0851r0>.